Supplementary Information for

A rising biocapacity total can mask ecological-capital land drawdown: a two-land model

Amin Abaee
Independent Researcher, Tehran, Iran
amin_abaee@ut.ac.ir ORCID 0000-0002-0019-1842

October 26, 2026

Equation numbers without the prefix S refer to the main text.

S1 Symbols and model extensions

A global hectare (gha) adjusts a hectare to world-average biological productivity for the relevant account year. Annual Ecological Footprint and biocapacity flows are reported in gha, with the annual basis implicit in that year (Global Footprint Network, 2009, 2019). In the model, B and E are annual account-equivalent quantities; in the empirical analysis, B and E_{prod} are observed NFA account values. Model contributions Y_f and Y_c and shortfall S use the same units, but are not separate NFA series. The debt state D is a leaky state driven by aggregate deficit $E - B$ and is measured in gha yr; $\dot{D}$ and ηD are in gha.

Table S1: Symbols, definitions, units, and roles. Per-hectare yields and contributions use the annual account basis (gha ha^{-1}). The coefficient $b_{G,c}$ is in gha yr ha^{-1}; annual account totals are in gha, with the accounting year implicit. The leaky debt state D is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality index	dimensionless	state
P	human population	cap	state
D	leaky-debt state driven by aggregate deficit $E - B$	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	OWID/FAOSTAT cereal yield, a physical productivity sentinel (not all crops)	tonnes ha^{-1}	empirical input
$\hat{b}_f(t)$	Base-anchored index-scaled proxy yield used only in SI S5	gha ha^{-1}	proxy construction

Symbol	Meaning	Unit	Role
b_{f0}	base fast-land yield per hectare	$\mathrm{gha\,ha^{-1}}$	parameter
b_c	capital-land direct yield	$\mathrm{gha\,ha^{-1}}$	parameter
$b_{G,c}$	capital regeneration-value coefficient	$\mathrm{gha\,yr\,ha^{-1}}$	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1-q/q_{\max})$	$\mathrm{yr^{-1}}$	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	$\mathrm{gha\,ha^{-1}}$	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	$\mathrm{gha\,ha^{-1}}$	derived
$Y_f,\ Y_c$	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived
B	model biocapacity $Y_f + Y_c$; or NFA total under its account convention	gha	aggregate
E_{prod}	NFA Global Ecological Footprint of Production	gha	empirical account
E	model-defined footprint demand, eP (or exogenous frozen demand)	gha	aggregate
$X,\ C$	crop-proxy account $X = \hat{b}_f A_f$ and residual $C = B - X$; neither is physical capital	gha	illustrative accounting split
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\mathrm{conv}}$	$\mathrm{ha\,yr^{-1}}$	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	$\mathrm{gha\,cap^{-1}}$	parameter
r	intrinsic population growth rate	$\mathrm{yr^{-1}}$	parameter
ρ_c	quality regeneration rate	$\mathrm{yr^{-1}}$	parameter
$q_{\max}$	quality ceiling	dimensionless	parameter
μ	cropland maintenance rate (reserve→fast)	$\mathrm{yr^{-1}}$	parameter
η_f	fast-land retirement rate (fast→reserve)	$\mathrm{yr^{-1}}$	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
η	debt-repayment rate	$\mathrm{yr^{-1}}$	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B-E]_+$ (reserve→capital)	$\mathrm{ha\,yr^{-1}}$	gated flow
χ_r	restoration coefficient	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1+e^{-\kappa_w(t-t_{wave})})$	$\mathrm{gha\,ha^{-1}}$	off except in SI S8
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
Δb, κ_w, t_{wave}	amplitude, steepness and center of the wave	$\mathrm{gha\,ha^{-1}}$; $\mathrm{yr^{-1}}$; yr	active only in SI S8
$A_c^{\min}$, $A_r^{\min}$, $A_f^{\max}$, $K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu+\eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived

Symbol	Meaning	Unit	Role
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0}kA_{tot} - (b_{f0}k - b_c)A_c$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	yr^{-1}	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-land shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$	dimensionless	derived
R_{fc}, χ	fast→capital restoration variant, $\chi A_f [B - E]_+$	$\text{ha}\,\text{yr}^{-1}$; $\text{gha}^{-1}\,\text{yr}^{-1}$	extension (off)
L_c, H_c	timber liquidation flow $[H_c - \sigma_c Y_c]_+ / b_{G,c}$ and its target	$\text{ha}\,\text{yr}^{-1}$; gha	extension (off)

Model extensions. The reported simulations do not include timber liquidation or direct restoration from fast land to capital land. SI §S6 considers reservations of the human-available shares as a policy extension. The technology wave is off in the baseline and gate-sign comparison; it is switched on only for the matched experiment in SI §S8.

Conventions. The positive-part operator is exact, conversion runs at the rate set by τ_{conv}, and transfers conserve area across A_f, A_c, and A_r. Quality q changes independently of area. Maintenance at rate μA_r allows fast-land area to remain positive at equilibrium. The reserve floor $A_r^{\min}$ and arable ceiling $A_f^{\max}$ are domain constraints; all reported trajectories stay within them, and reserve land remains at least $0.80\,ha$ in the drawdown sweeps. We also enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, and $K \geq K_{\min}$.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiate $B = b_f A_f + \Upsilon_c(q) A_c$, use $b_f = (b_{f0} + T_b)e^{-\alpha D}$, and substitute Eqs. (6)–(7) to obtain the general identity

$$\dot{B} = \Delta b_{\text{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c} G_c'(q) A_c\, \dot{q} + \Upsilon_c R_{rc} - \alpha \dot{D}\, b_f A_f + \frac{\dot{T}_b}{b_{f0} + T_b}\, b_f A_f. \quad \text{(S1)}$$

The conversion contribution has the sign of Δb_{conv}. The quality term follows the sign of $G_c'(q)$: it is non-negative for $q \leq q_{\max}/2$ and non-positive above that point, since $\dot{q} \geq 0$. While debt accumulates, its contribution is non-positive; a rising technology wave and restoration contribute non-negatively. Proposition 1 in the main text sets $T_b \equiv 0$, $R_{rc} \equiv 0$, and holds D fixed. In the demonstration run, D instead rises to $0.20\,gha\,yr$ and then declines. Initially, conversion contributes $0.21\,\text{gha}\,\text{yr}^{-1}$ to $\dot{B}$, compared with $-0.005\,\text{gha}\,\text{yr}^{-1}$ from fast-land maintenance and retirement and a debt contribution of about $-0.004\,\text{gha}\,\text{yr}^{-1}$. Conversion therefore dominates the initial change in aggregate model biocapacity.

S2.2 The deficit-side affine system and its spectrum

On the deficit side, set $q = q_{\max}$, freeze D, switch off R_{rc}, and assume $S/b_f < A_c - A_c^{\min}$. Equation (5) then gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$. Using $A_r = A_{tot} - A_f - A_c$, Eqs. (6)–(7) reduce to

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \tag{S2}$$

The matrix has trace $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and determinant $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$, where $k = \mu/(\mu + \eta_f)$. At baseline, $\operatorname{tr} J = -1.0512\,\text{yr}^{-1}$ and $\det J = 0.05353\,\text{yr}^{-2}$. Its eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\text{yr}^{-1},\ -0.9975\,\text{yr}^{-1}. \tag{S3}$$

Because J depends on neither demand E nor the current state, this spectrum stays constant along the frozen-demand branch. At $E/E_{\text{ceil}} = 0.50,\ 0.90,\ 0.99,\ 0.999$, the slow eigenvalue remains $\lambda_1 = -0.05366\,\text{yr}^{-1}$. It is unchanged as ρ_c varies ($\lambda_1 = -0.0537\,\text{yr}^{-1}$ at $\rho_c = 0.04,\ 0.08,\ 0.16\,\text{yr}^{-1}$), shifts modestly with τ_{conv} ($-0.0553,\ -0.0537,\ -0.0505,\ -0.0417\,\text{yr}^{-1}$ at $\tau_{\text{conv}} = 0.5,\ 1,\ 2,\ 5\,\text{yr}$), and shifts more with μ ($-0.0254,\ -0.0537,\ -0.1099\,\text{yr}^{-1}$ at $\mu = 0.03,\ 0.06,\ 0.12\,\text{yr}^{-1}$). It remains negative in every tested sweep.

The affine system's fixed point is $A_c^*(E)$ from Eq. (18), with $A_f^* = k(A_{tot} - A_c^*)$. For $E > E_{\text{ceil}}$, this point lies below $A_c^{\min}$. The trajectory instead approaches the boundary $A_c = A_c^{\min}$ asymptotically, with $u_c \to 0$ and persistent shortfall $S^* = E - E_{\text{ceil}}$. At baseline, two members of the no-conversion equilibrium family are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\text{ha}, 0.824\,\text{ha}, 0.762\,\text{ha}, 1.488\,\text{cap})$ and $(1.091\,\text{ha}, 0.500\,\text{ha}, 0.909\,\text{ha}, 1.731\,\text{cap})$. Both satisfy $A_r^*/A_f^* = \eta_f/\mu = 0.8333$; the second is a sustainable cropland equilibrium supported by maintenance. This reserve-share mode explains why the strict-inequality interval in the delayed-quality demonstration lasts about 149 yr, even though most conversion is complete within 25 yr.

When the other branch of the minimum applies ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$. The distance to the floor then follows $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it shrinks exponentially but never reaches zero in finite time. The floor is approached asymptotically, so operational proximity thresholds depend on the tolerance used to define "near" (Section S4.2).

S2.3 Characteristic structure of the full delay system

Let $z = (A_f, A_c, q, P, D)$. On any fixed active set, the linearized delay system is $\dot{z}(t) = J_0 z(t) + J_g z(t - \tau_g) + J_p z(t - \tau_p)$, with characteristic equation $\det \Delta(\lambda) = 0$ and $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda \tau_g} - J_p e^{-\lambda \tau_p}$. The current-state matrix J_0 contains the conversion and debt feedbacks: the 2×2 land block above, the deficit-side debt row $\partial \dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c} G_c' A_c, e, -\eta)$, and the couplings $\partial b_f/\partial D = -\alpha b_f$. The regeneration-delay matrix J_g has the single entry $G_c'(q^*)$ in the q row. The population-delay matrix J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position, from $\partial \dot{P}/\partial K = rP^2/K^2$ and $K = B/e$.

At baseline, two scalar delayed modes decouple. Linearized at $q_{\max}$, the quality mode is $\dot{\xi} = -\rho_c \xi(t - \tau_g)$, with undelayed rate $-0.08\,\text{yr}^{-1}$ and characteristic equation $\lambda = -\rho_c e^{-\lambda \tau_g}$. Since $\rho_c \tau_g = 1.6 > \pi/2$, this linearized mode would oscillate. The nonlinear model cannot follow that oscillation: $\dot{q} = G_c(q(t - \tau_g)) \geq 0$ for $q(t - \tau_g) \in [0, q_{\max}]$, and q is clamped at $q_{\max}$. Quality is therefore non-decreasing and converges to $q_{\max}$; the oscillation is an artifact of linearizing a one-sided law at its boundary. On the no-conversion face, the population mode follows the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at baseline.

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\text{ceil}}$, capital-land area approaches $A_c = A_c^{\min}$ and the persistent shortfall feeds debt. At the floor, $B_{eq}(A_c^{\min}; D) = b_{f0}e^{-\alpha D}k(A_{tot}-A_c^{\min})+b_cA_c^{\min}$, so $\dot{D} = E-B_{eq}(A_c^{\min}; D)-\eta D$. Its fixed point $D^*(E)$ satisfies $\eta D = E - B_{eq}(A_c^{\min}; D)$. The right-hand side is increasing in D, making the fixed point unique and attracting when $\eta > \alpha b_{f0}k(A_{tot} - A_c^{\min})$; at baseline, $0.034\,\text{yr}^{-1} < 0.05\,\text{yr}^{-1}$. Baseline values are $D^* = 1.186\,\text{gha yr}$ at $E = 1.158\,\text{gha}$ and $D^* = 5.375\,\text{gha yr}$ at $E = 1.238\,\text{gha}$. Corresponding $B(D^*)$ values are 1.099 and 0.969 gha, with persistent shortfalls $S^* = 0.059$ and 0.269 gha. Under fixed demand, debt therefore saturates; it is unbounded only if the shortfall is unbounded.

S2.5 Boundedness asymmetry and the crossover shortfall

Equation (12) gives $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The technology term is bounded between $\ln b_{f0}$ and $\ln(b_{f0} + \Delta b)$, so its maximum contribution to $\ln b_f$ is $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ at baseline. From Eq. (13), $\dot{D} \le \bar{S} - \eta D$, where $\bar{S}$ is the running supremum of shortfall. Thus $D(t) \le \max\{D(0), \bar{S}/\eta\}$, and the debt channel contributes no less than $-\alpha\bar{S}/\eta$. The channels are equal in magnitude at $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\text{gha}$.

The fast-land contribution $Y_f = b_f A_f$ has the same two-channel structure; capital contribution $\Upsilon_c A_c$ is unaffected by either. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, the effects are directly comparable as bounded increments (Δb versus αD^*), so the logarithmic asymmetry disappears. The bound above applies to the multiplicative specification.

S2.6 Macro-ratio identities

Set $R_B = E/B$, $R_A = E/Y_f$, and $\psi_f = Y_f/B$. Then $R_A = R_B/\psi_f$ by definition. The direct-yield share of capital contribution, $\psi_c = b_cA_c/Y_c$, is a different ratio; $R_A = R_B/\psi_c$ is not an identity. Aggregate balance occurs at $R_B = 1$. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ is necessary but not sufficient for conversion: shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$. The threshold changes when flows are reserved. Capital-land area falls when $u_c > R_{rc}$, with u_c defined in Eq. (5). Thus, under baseline shares, conversion requires $R_B > 1$. Under the reservation extension with $\sigma_f, \sigma_c < 1$, conversion and declining A_c can occur even when $R_B \le 1$; aggregate overshoot is not required. At $q^* = q_{\max}/2$, the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at baseline. This value applies at that point; it is not a structural constant.

S3 Numerical methods and sensitivity analyses

Numerical method. We integrate the delay equations by the method of steps with an explicit first-order Euler scheme. The positive-part operator is exact. We enforce $A_c \ge A_c^{\min}$, $A_f \ge 0$, $q \in [0, q_{\max}]$, $P, D \ge 0$, and $K \ge K_{\min}$, and calculate reserve area as $A_r = A_{tot} - A_f - A_c$. Histories of B and q remain at their initial values over $[-\max(\tau_g, \tau_p), 0)$. In the technology-wave experiment, both runs share the same no-wave history. The active $T_b(0)$ changes current $B(0)$, and later wave-adjusted values enter the delayed input $K(t - \tau_p)$. We use $\Delta t = 0.05\,\text{yr}$ for the demonstration and threshold bisection, and 0.1 yr for drawdown sweeps; selected thresholds are checked at 0.05 yr. At $A_{c0} = 1.0\,\text{ha}$, the 10^{-6} ha floor-proximity value shifts from 2.955 to 2.98 cap across these step sizes. We therefore report thresholds as bands that depend on tolerance and numerical resolution.

Delayed versus current quality in the regeneration term. All reported trajectories use the delayed law $G_c(q(t - \tau_g))$ in Eq. (10). We also test the current-quality alternative, $G_c(q(t))$.

Because q is monotone and regeneration contributes at most $0.016\,\mathrm{gha\,ha^{-1}}$—compared with direct yield $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$—the summary differences stay below 1%. The current-quality version extends the demonstration window by 1.3 yr and shifts the initial-population floor-proximity value by at most 0.001 cap (Table S2).

Table S2: Sensitivity to evaluating regeneration at current rather than delayed quality. The baseline parameter set and $\tau_p = 25\,\mathrm{yr}$ are used; the reported trajectories use the delayed-quality specification. Biocapacity B, fast-land contribution Y_f, and capital-land contribution $\Upsilon_c(q)A_c$ are in gha; land areas A_c, A_f, A_{c0} are in ha; the operational initial-population floor-proximity threshold P_0 is in cap; delays and step intervals are in yr.

Quantity	current $q(t)$	delayed $q(t-\tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial → mask end, gha)	0.08364 → 0.04038	0.08364 → 0.04038
Fast-land contribution Y_f (initial → mask end, gha)	0.51000 → 0.78462	0.51000 → 0.78462
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 5\,\mathrm{yr}$, $\Delta t = 0.1\,\mathrm{yr}$)	2.955	2.955
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 20\,\mathrm{yr}$)	2.955	2.954
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 40\,\mathrm{yr}$)	2.955	2.954

Well-posedness. Under the stated bounds, the admissible state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$ is forward invariant. The positive-part and minimum operators are Lipschitz, giving a unique delay-system solution for each admissible history. We evaluate stability at $S = 0$ from the deficit side.

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 extends the main-text fast-land-yield sweep in Table 2(a) by reporting the balance-branch slope. The sign of Δb_{conv} stays positive when $b_{G,c}$ triples. At $b_f = 0.85\,\mathrm{gha\,ha^{-1}}$, $\Upsilon_c(q^*) = 0.058, 0.066, 0.074\,\mathrm{gha\,ha^{-1}}$ for $b_{G,c} = 0.4, 0.8, 1.2\,\mathrm{gha\,yr\,ha^{-1}}$, giving $\Delta b_{\mathrm{conv}} = +0.792, +0.784, +0.776\,\mathrm{gha\,ha^{-1}}$.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 reports the minimum capital-land area reached in endogenous-demand runs over a grid of initial populations P_0. The runs use $A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25\,\mathrm{yr}$, $\chi_r = 0.10$, $\Delta t = 0.1\,\mathrm{yr}$, and $T = 400\,\mathrm{yr}$; the full P_0 grid (step 0.1) is in `si_drawdown_depth.csv`. Depending on P_0, the minimum occurs 3–25 yr after the run begins, when the shortfall first closes ($B \geq E$).

At $A_{c0} = 1.00$ ha, the delayed-quality 10^{-6} ha proximity value varies by at most 0.001 cap across $\tau_g = 5, 20, 40$ yr (SI Table S2 in § S3); that comparison covers only the tested range. Drawdown depth varies continuously and monotonically with both P_0 and A_{c0}.

The initial populations at which the minimum comes within 10^{-2}, 10^{-3}, and 10^{-4} ha of the floor are 1.8, 1.9, 2.1 cap for $A_{c0} = 0.40$ ha; 2.2, 2.3, 2.4 for 0.70 ha; 2.6, 2.7, 2.8 for 1.00 ha; and 2.8, 2.9, 3.0 for 1.18 ha. Bisection with a 10^{-6} ha proximity tolerance ($\Delta t = 0.05$ yr, $T = 1000$ yr) gives 2.41, 2.70, 2.98, 3.14 cap. The thresholds match to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.955 cap at $A_{c0} = 0.40$ and 1.00 ha, $\Delta t = 0.1$ yr).

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05$ ha.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

We measure recovery as $\mathcal{R}_c(T) = (A_c(T) - A_c^{\deg})/(A_c^{\text{init}} - A_c^{\deg})$. A value of 0 means no recovery, 1 means return to the reference, a negative value means further decline, and a value above 1 means over-restoration. Table S5 applies this metric to delayed-quality runs with $A_{c0} = 1.0$ ha ($A_c^{\text{init}} = 1.0$ ha, $A_c^{\deg} = \min_t A_c$, $T = 400$ yr). Without restoration, A_c stays at its minimum because no return flow exists. With surplus-gated restoration, recovery is partial in all six initial-population cases ($\mathcal{R}_c = 0.21$–0.89); none returns to the reference. The separate gate-sign experiment in SI §S4 uses other initial states and interventions, including cases of over-restoration.

Table S5: Recovery of the capital-land area after a demand-driven drawdown ($A_{c0} = 1.0$ ha, $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $T = 400$ yr): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	0.969	0.89
2.0	1.100	0.398	3.8	0.398	0.677	0.46
2.5	1.375	0.075	4.1	0.075	0.433	0.39
3.0	1.650	0.050	13.8	0.050	0.328	0.29
3.5	1.925	0.050	20.6	0.050	0.278	0.24
4.0	2.200	0.050	26.6	0.050	0.250	0.21

S4.4 Gate-sign phase dependence

The gate-sign comparison starts from a post-drawdown state: $A_c = 0.058$ ha (near the floor), $A_f = 0.6$ ha, $q = 0.9$, and $P = 1.2$ cap. At the degraded equilibrium, $R_B = 1$ and $B = E$, so neither restoration gate is active. The reference equilibrium has $A_c^{\text{init}} = 0.8225$ ha and $P^{\text{init}} = 1.489$ cap; the degraded reference is $A_c^{\deg} = 0.0513$ ha. Every row uses $T_b = 0$ and the same starting state.

Table S6: Restoration gate and recovery outcomes at $T = 100$ and 900 yr. The table reports end-state land areas, population relative to the reference, trajectory extrema, and peak debt D (gha yr). Annual account values $B = E$ are in gha; land areas are in ha; population ratios and recovery metrics are dimensionless. Each row approaches $R_B = 1$ ($E = B$).

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.097	0.765	0.638	0.705	0.86	0.602	1.097	0.543
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Without restoration, A_c stays at its minimum because the model has no return flow. Surplus-gated restoration produces partial or over-restoration depending on χ_r. At $\chi_r \geq 0.05$, it continues while $B > E$, and supported population falls to 0.97–0.90 of its reference value. A target cap returns A_c to its reference ($\mathcal{R}_c = 1.00$). Deficit-gated restoration acts during shortfall, but at the tested rates it is weaker than concurrent conversion, leaving recovery near zero. Freezing conversion under surplus-gated restoration raises peak debt from 0.152 to 0.543 gha yr; restoration then shifts reserve land to capital use while fast-land maintenance declines. These outcomes depend on the starting state and intervention rates. The gate-sign comparison keeps $T_b = 0$; SI §S8 reports the separate matched wave/no-wave debt experiment.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutrality	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at the conversion margin; conversion alone cannot produce a composition mask

S5 Empirical details

S5.1 Data and vintages

Our empirical source is the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The workbook spans 1961–2025, but we use complete world observations through 2023 and exclude the estimated 2024–2025 rows. The 1961–2022 sensitivity uses the same edition and population series. This release recalculates historical values and reports five biocapacity account categories: fishing grounds, built-up land, cropland, grazing land, and forest biocapacity. These categories are account entries, not a common-boundary inventory of physical stocks.

The workbook's `pop_1000_persons` field reports UN PopStat population in thousands; we convert it to persons before calculating per-capita values. The field `efp_total_gha` gives the Global Ecological Footprint of Production, denoted E_{prod}. We calculate per-capita values from unrounded account totals and population rather than the workbook's rounded columns. Table S8 summarizes the endpoints. From 1961 to 2023, B rises from 9,725,417,562 to 12,142,843,961 gha, population from 3.06487 to 8.09173 billion, and B/P falls from 3.17319 to 1.50065 gha person^{-1}. Over the same period, E_{prod}/P rises from 2.19700 to 2.56462 gha person^{-1}. The identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ holds to rounding; it separates extensive and intensive changes but does not attribute causes. In both windows, 1973 is the first year with $E_{\text{prod}}/B > 1$.

Table S8: World endpoint changes in NFA Edition 2026. The 1961–2022 column is a same-edition sensitivity; 2024–2025 estimates are excluded. Population and per-capita quantities are recomputed from workbook totals.

Measure	1961–2023	1961–2022
$\Delta B/B_{1961}$	+24.8568%	+24.5450%
$\Delta \ln B$	+0.221997	+0.219497
$\Delta P/P_{1961}$	+164.0154%	+161.7211%
$\Delta \ln P$	+0.970837	+0.962109
$\Delta(B/P)/(B/P)_{1961}$	−52.7085%	−52.4131%
$\Delta \ln(B/P)$	−0.748840	−0.742612
$\Delta(E_{\text{prod}}/P)/(E_{\text{prod}}/P)_{1961}$	+16.7327%	+17.6994%
First year $E_{\text{prod}}/B > 1$	1973	1973

S5.2 Divisia decomposition, sensitivity to the base-year share and level fallback

The crop proxy is an illustrative account construction. We combine independently sourced OWID/FAOSTAT world cropland area, defined as arable plus permanent-crop area (ha), with cereal yield (tonnes per hectare) (Our World in Data, 2026b,a). Cereal yield is a productivity sentinel, not a measure of output from all crops. For a declared 1961 account-share anchor ϕ_0, we set $\hat{b}_f(t) = \hat{b}_f(1961)\tilde{b}_f(t)/\tilde{b}_f(1961)$, with $\hat{b}_f(1961)A_f(1961) = \phi_0 B(1961)$. We then define $X_t = \hat{b}_f(t)A_f(t)$ and $C_t = B_t - X_t$. Thus X is an index-scaled account proxy, and C is the residual that reconciles it to the NFA total. The residual is not observed ecological capital, non-cropland biocapacity, or the model quantity $\Upsilon_c(q)A_c$.

When both X and C are positive at the two endpoints, the exact log-mean Divisia identity is

$$\Delta \ln B = w_X\left(\Delta \ln \tilde{b}_f + \Delta \ln A_f\right) + w_C \Delta \ln C, \qquad w_X = \frac{L(X_T, X_0)}{L(B_T, B_0)}, \quad w_C = \frac{L(C_T, C_0)}{L(B_T, B_0)}, \tag{S4}$$

Here $L(a, b) = (b - a)/(\ln b - \ln a)$ is the logarithmic mean, with its continuous limit when $a = b$. We use the measured 1961 NFA cropland share, $\phi_0 = 0.1286472202$, as the central anchor. Mapping that account share to the external area–yield proxy is an assumption, not an independent observation of physical capital. Table S9 gives the selected sensitivity for 1961–2023. At the central anchor, the cereal-yield, area, and residual terms are +0.268662, +0.037710, and −0.084375, summing to $\Delta \ln B = +0.221997$. The same-edition 1961–2022 sensitivity gives +0.264696, +0.037506, and −0.082705, summing to +0.219497. The decomposition is exact for this proxy, but its allocation depends on ϕ_0 and is not a physical-stock attribution.

Table S9: Sensitivity of the proxy Divisia decomposition to the 1961 account-share anchor ϕ_0 (world, 1961–2023). Contributions and weights are dimensionless. The $\phi_0 = 0.128647$ row uses the measured NFA cropland share in 1961; other rows are illustrative calibrations.

ϕ_0	w_X	w_C	Cereal yield	Area	Residual C	Sum
0.100000	0.1833	0.7957	+0.2088	+0.0293	−0.0162	+0.2220
0.128647	0.2358	0.7352	+0.2687	+0.0377	−0.0844	+0.2220
0.150000	0.2749	0.6893	+0.3133	+0.0440	−0.1352	+0.2220
0.190000	0.3483	0.6008	+0.3968	+0.0557	−0.2305	+0.2220
0.250000	0.4582	0.4581	+0.5221	+0.0733	−0.3734	+0.2220
0.300000	0.5499	0.3178	+0.6265	+0.0879	−0.4924	+0.2220
0.320000	0.5866	0.2454	+0.6683	+0.0938	−0.5401	+0.2220
0.340000	0.6232	0.1009	+0.7100	+0.0997	−0.5877	+0.2220

For $\phi_0 \geq 0.340531$ in 2023 (or 0.343400 in 2022), terminal residual C is nonpositive, so logarithmic weighting is invalid. We use the exact level identity $\Delta B = \Delta X + \Delta C$ instead; this is still an accounting identity, not a stock measure. At the selected anchor, $X_{2023}/B_{2023} = 0.377784$ and C remains positive, but the proxy fast-land share and residual contribution vary across the sensitivity analysis. Every row uses the same NFA total B. The residual absorbs differences between the index-scaled crop proxy and that total, including account categories, boundaries, valuation, vintage, and calibration.

S5.3 Cropland-account calibration

We report NFA cropland-account movements separately from the illustrative proxy. From 1961 to 2023, NFA cropland biocapacity rises 2.870055-fold and NFA cropland area rises 1.198029-fold; their account-value-per-area quotient rises 2.395647-fold. For 1961–2022, the same-edition ratios are 2.851441, 1.198326, and 2.379520. This quotient is an NFA account value per area, not an independent physical crop-yield measure. Over 1961–2023, the external OWID/FAOSTAT area series rises 1.173414-fold and cereal yield rises 3.124667-fold. Those independent series serve as

a sentinel proxy; they do not reconstruct the NFA cropland account or measure output from all crops. The model's representative $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$ remains a conditional parameter, not an empirical fit to these ratios.

S5.4 Observation-operator hierarchy

- *NFA accounting observations.* Total and component values in gha describe NFA account composition and change under the stated edition and conventions. They do not directly measure biomass, soil condition, wetland extent, fish biomass, or model state A_c.
- *Fast-provisioning area.* The external world cropland-area series covers arable land plus permanent crops for 1961–2023 (ha) (Our World in Data, 2026b). It is an independent proxy input, not the NFA cropland-area series.
- *Crop-productivity sentinel.* OWID/FAOSTAT cereal yield is reported in tonnes per hectare for 1961–2024 (Our World in Data, 2026a). It covers cereals, not all crops, and is not a physical-stock measure. Base-anchored $\hat{b}_f$ is used only in the illustrative account proxy.
- *Ecological-capital components.* Separate physical observation operators could use forest extent and biomass, soil carbon and condition, wetland extent and condition, and fish biomass by species or stock. Such data would need harmonized boundaries, reference years, definitions, uncertainty, and explicit treatment of overlap. We assembled no common-boundary global physical-stock series for these components. NFA forest biocapacity and residual C cannot substitute for those measurements.
- *Model mapping.* The model defines capital contribution as $Y_c = \Upsilon_c(q)A_c$. Neither A_c nor q is identified empirically by total B or an NFA component, and we do not assume proxy residual $C = B - X$ equals Y_c. A model–data comparison would first need independent physical measures and an explicit mapping.

Table S10 summarizes the possible links while keeping NFA categories distinct from model states.

Table S10: Conceptual mapping from two-land model quantities to NFA information and independent proxy inputs. The rows specify possible observation operators, not identities between NFA categories and model states. Annual account values are in gha; land areas are in ha.

Model quantity	Closest account or proxy information	Interpretation and identification limit
Fast area and proxy account A_f, $X = \hat{b}_f A_f$	NFA cropland account; independently sourced OWID cropland area and cereal yield	NFA cropland biocapacity is an account value; the external area–yield product is an illustrative, base-anchored proxy. Neither is an observed model fast-land yield or a one-to-one physical measure.
Capital area, quality and contribution A_c, q, $Y_c = \Upsilon_c(q)A_c$	No direct NFA counterpart; the proxy residual $C = B - X$ is defined by subtraction	In the model $Y_c = \Upsilon_c(q)A_c$; empirically C is not assumed equal to Y_c. It also absorbs differences in boundaries, account categories (including built-up land), valuation, vintage and proxy calibration, and is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S11: Observability ladder: what is directly measured, what is identifiable by construction, and what remains unidentified.

Tier	Quantities	Status
Observable physical area input	OWID cropland area [ha]; other mapped land-cover classes where available	Measured extent, but mapping to effective A_c is not direct
Observable population	P [persons]	Direct in the UN PopStat workbook field after unit conversion
Observable productivity sentinel	Cereal yield [tonnes ha^{-1}]	Physical yield of cereals only; not all-crops output or ecological capital
Observable annual account value	NFA B, E_{prod} and components [gha]	Measured only under the NFA accounting convention; value-weighted accounts
Proxy account and residual	$X = \hat{b}_f A_f$ and $C = B - X$ [gha]	Identified by construction for the stated inputs and anchor; not physical stocks and not assumed equal to $\Upsilon_c(q)A_c$
Effective capital contribution	$Y_c = \Upsilon_c(q)A_c$ [gha]	A model quantity; no common-boundary physical observation operator is assembled here
Not separately identifiable	A_c, q, b_c, $b_{G,c}$, ρ_c and D	Requires independent physical-state, quality, regeneration or degradation observations and a declared mapping

S5.5 Regeneration anchors

The model's capital-land regeneration timescale represents a lumped recruitment response, not a universal ecological rate. In Neotropical secondary forests, plot-level age and above-ground biomass records (2ndFOR; 1,334 plots, 41 chronosequences, 35 good fits) give median $t_{50} \approx 30$ yr and $\tau_r \approx 38$ yr under a lagged saturating recovery model (Poorter et al., 2016). For 33 fish stocks classified as collapsed in RAM Legacy, trajectories of B/B_{MSY} give $t_{50} \approx 5.6$ yr and $\tau_r \approx 3.4$ yr (Hutchings and Reynolds, 2004; Neubauer et al., 2013). Soil organic carbon re-equilibrates over about 17–23 yr after land-use change (Poeplau et al., 2011). Forest biomass reaches about half of old-growth levels by 20 yr and 90% at a median of about 66 yr; roughly 29% of collapsed stocks recover to 50% within 5–15 yr.

These observations describe different systems and are not pooled into one rate. Forest and soil records provide decadal-scale context for the representative effective response scale of about 29 yr. The 10–40 yr interval is a modeling range for the lumped capital-land class, not a pooled empirical estimate. The faster fish-stock episodes are not treated as a lower bound for that class. Neither ρ_c nor delay τ_g is calibrated from these records.

S5.6 Recursive identifiability in the land-type accounts

NFA Edition 2026 illustrates why the aggregate should be read with its components, without treating account movements as physical-stock changes. From 1961 to 2023, cropland contributes +24.0577 percentage points to the +24.8568% change in total B; built-up land adds +7.4416, while Forest biocapacity contributes −5.9525, grazing land −0.2688, and fishing grounds −0.4212. The non-cropland sum, which includes built-up land, changes by only +0.7990 points (1.0092×) because unlike account movements offset one another.

For the same-edition 1961–2022 sensitivity, contributions are +23.8183 points for cropland, +7.2854 for built-up land, −5.8137 for Forest biocapacity, −0.2890 for grazing land, and −0.4559 for fishing grounds. The non-cropland sum changes by +0.7267 points (1.0083×). These are accounting attributions, not evidence that forest biomass, soil condition, wetland extent, fish biomass, or model A_c declined. Built-up land is an NFA account class, not ecological capital; including it in the non-cropland sum is arithmetic, not an ecological aggregation.

S5.7 Physical-stock scoreboard: specification, not inventory

We assembled no common-boundary global physical-stock series for these components, so we present no physical-stock composite or global decline finding. This does not mean that such a series could never be assembled. Table S12 is an unpopulated specification of candidate measures and comparability requirements; it contains no empirical endpoints and is not a model calibration.

These physical measures are heterogeneous. Forest biomass and soil carbon are masses; wetland and cropland extent are areas; fish biomass is stock-specific; freshwater storage is a volume; and condition indicators are often dimensionless or categorical. They are not dimensionally comparable and may overlap spatially. Annual flows—harvest, catch, increment, recharge, or emissions—must also remain distinct from stocks and extents. Even after normalization, interpretation depends on overlap, baselines, uncertainty, ecological thresholds, valuation, and allowed substitution. Unless common spatial and temporal boundaries, a normalization or valuation rule, and substitution rules are specified, the components should remain a nonadditive vector. A scalar composite is not neutral and cannot be inferred from NFA gha accounts. SEEA Ecosystem Accounting likewise distinguishes ecosystem extent and condition from ecosystem-service and monetary presentations (United Nations et al., 2021).

In the two-land model, A_c is an effective area state (ha) and q is a separate quality index; neither is observed in the NFA total. Physical-stock data could inform a model observation operator only after boundaries and mappings to these effective states are specified. Proxy residual

Table S12: Physical-stock scoreboard specification. The rows are unpopulated: no common-boundary global physical-stock series for the relevant components or endpoint values were assembled. Stocks, extents, condition measures, and annual flows are not combined into a composite.

Component	Physical stock, extent, or condition to define	Annual flow or pressure (separate)	Comparability rule required
Forests	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition	Net annual increment, harvest, or deforestation	Common forest definition and mapped area; reference year; treatment of plantations and overlap
Soils	Soil organic-carbon stock or density for a stated depth; structure or erosion condition	Carbon accumulation or loss; erosion where measured	Consistent depth, bulk density, land cover, and sampling method
Wetlands	Extent (ha) and an explicit ecological-condition measure	Carbon accumulation, water-storage change, or nutrient flux	Wetland definition and overlap with forest and other land-cover classes
Fisheries	Spawning-stock biomass by species or stock (tonnes); age or size structure	Recruitment, catch, and landings	Stock identity, assessment method, reference point, and uncertainty
Freshwater	Surface-water or groundwater storage (m^3), level, and quality	Recharge and withdrawal	Basin or aquifer boundary; separate stored volume from annual flow

$C = B - X = B - \hat{b}_f A_f$ is not ecological capital, a measured non-cropland stock, or an estimate of $\Upsilon_c(q)A_c$. NFA components remain account values, not physical-stock observations.

S6 Policy extensions

Reserved biocapacity share. If both land classes have the same reservation share, $\sigma_f = \sigma_c = \sigma < 1$, human-available supply is σB and shortfall is $S = [E - \sigma B]_+$. Reservation lowers the demand at which conversion begins; conversion is off while $E \le \sigma B$. With different shares, the condition becomes $E \le \sigma_f Y_f + \sigma_c Y_c$, which need not equal σB. These results apply to the stated parameterization; they do not establish robustness to disturbances or parameter uncertainty.

Typed floors and conversion stop. The model imposes separate constraints on reserve land ($A_r \ge A_r^{\min}$), capital land ($A_c \ge A_c^{\min}$), fast land ($A_f \le A_f^{\max}$), and carrying capacity ($K \ge K_{\min}$). A stop rule triggered at $A_c = A_c^{\min}$ preserves the capital-land floor. A ceiling on A_f limits fast-land extent but does not ensure that A_c stays above its floor unless the stop rule also accounts for the capital floor and remaining-area constraints. In Eq. (5), conversion slows to zero as $A_c \to A_c^{\min}$. A control based only on total B cannot guarantee this typed floor because B does not identify A_c.

Institutional signature. Related work distinguishes protective and mobilizing institutional feedback (Abaee, 2026a). Here, surplus-gated restoration is off during shortfall, while deficit-gated restoration acts during shortfall and can further reduce provision (Section S4.4).

Aggregation and typed feasibility. A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Sections 2 and 4 of Abaee, 2026b). ECOMOD's B is also a linear sum, but the companion considers a separate finite-plan setting. There, the compensatory protocol accepts states even when no available plan meets every component floor (Theorem 2 in Abaee (2026b)). Theorem 4(iii) in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe. Thus no positive-elasticity member is uniformly safe on that region. Under pathwise feasibility, falling below a typed floor during transition remains a violation even if the endpoint later recovers. The comparison distinguishes an aggregate criterion from constraints on individual components.

S7 The one-stock comparator

We include this one-stock formulation as a structural comparator, not as a reduction of the two-land model. Its state A has comparator-specific stock units and is distinct from two-land state A_c. The equations are

$$
\begin{aligned}
&G(A) = \rho A\left(1 - \frac{A}{A_{\max}}\right), \qquad B(t) = b(t)A(t) + b_G G(A(t)), \qquad E(t) = eP(t);\\
&\dot{A}(t) = G\big(A(t-\tau_g)\big) - \frac{[E(t) - \sigma b(t)A(t)]_+}{b_G};\\
&\dot{P}(t) = rP(t)\left(1 - \frac{P(t)}{K(t-\tau_p)}\right), \qquad K(t) = \max\left(\frac{B(t)}{e}, K_{\min}\right);\\
&\dot{D}(t) = [E(t) - B(t)]_+ - \eta D(t), \qquad b(t) = \big(b_0 + T_b(t)\big)e^{-\alpha D(t)};\\
&T_b(t) = \omega_{Tb}\frac{\Delta b}{1 + \exp[-\kappa(t - t_w)]}.
\end{aligned}
$$

Here σ is the dimensionless human-available share; $\omega_{Tb} \in \{0, 1\}$ switches the logistic technology increment on or off. The states A, $A_{\max}$, and imposed floor A_{ext} use comparator-specific stock units. $A_{\max}$ sets the logistic growth scale; it is not an imposed upper bound. Accordingly, $G(A)$

is in stock units per year. The coefficients b, b_0, T_b, and Δb are in gha per stock unit, while b_G is in gha yr per stock unit, making both terms in B annual account values in gha.

B and E use the GFN/NFA convention, with the annual basis implicit. Population P, carrying capacity K, and $K_{\min}$ are in cap; e is in gha cap^{-1}. Debt D is in gha yr, so $\dot{D}$ and ηD are in gha, and α is in gha^{-1} yr^{-1}. Rates ρ, r, η, κ are in yr^{-1}; delays τ_g, τ_p and wave midpoint t_w are in years.

The numerical implementation enforces $A \geq A_{\text{ext}}$, $P, D \geq 0$, and $K \geq K_{\min}$, with the exact positive-part operator. Delayed histories are constant at their initial values, and RK4 integration uses linear interpolation of delayed states. Step sizes are 0.1 yr for the full trajectory (checked at 0.05 and 0.025 yr), 0.01 yr for the fixed-demand mask (with a 0.02 yr shortfall sweep), 0.2 yr for the delay pair (checked at 0.1 yr), and 0.5 yr for the finite-grid classification.

With fixed b, $T_b = 0$, $D = 0$, and $\sigma = 1$, interior equilibria satisfying $0 < A < A_{\max}$ and $B(A)/e > K_{\min}$ obey $E = B(A)$ and $P = B(A)/e$. They form a one-parameter family along

$$B(A) = (b + b_G\rho)A - \frac{b_G\rho}{A_{\max}}A^2, \qquad B'(A) = b + b_G\rho - 2b_G\rho\frac{A}{A_{\max}}.$$

An interior maximum within $0 < A < A_{\max}$ requires $b_G\rho > b$. For the debt-free, constant-parameter delay and basin illustrations, $b = 0.5$, $b_G = 0.8$, and $\rho = 0.05$, so $b_G\rho = 0.04 < b$ and $B(A)$ increases throughout that interval. In the full-trajectory illustration, $b(t)$ changes with the technology wave and debt; we make no monotonicity or equilibrium-fold claim for that time-varying case.

For the debt-free reference equilibrium, $A^* = 0.8$ under $(b, b_G, \rho, A_{\max}, e, r) = (0.5, 0.8, 0.05, 1.2, 0.55, 0.02)$. It has $B^* = 0.4107$ gha and $P^* = B^*/e = 0.7467$ cap. The characteristic equation has a neutral root tangent to the equilibrium family and a positive real root. That positive root is $0.5883\,\text{yr}^{-1}$ when $(\tau_g, \tau_p) = (0, 0)$ and $0.6250\,\text{yr}^{-1}$ when $(30, 25)$ yr. These local results apply only to the comparator; they are neither a Hopf classification nor a stability result for the two-land system.

Figures S1–S4 show conditional runs of the comparator. In Fig. S3, the same starting state approaches $A_{\max}$ at $\tau_g = 10$ yr but reaches the imposed floor at $\tau_g = 30$ yr, with $\tau_p = 25$ yr in both cases. This pair shows delay sensitivity for one initial condition; it does not establish a universal critical delay. Figure S4 classifies a finite grid of starting states at fixed τ_p. Grid-cell proportions are not probabilities over initial conditions or measures of a continuous basin.

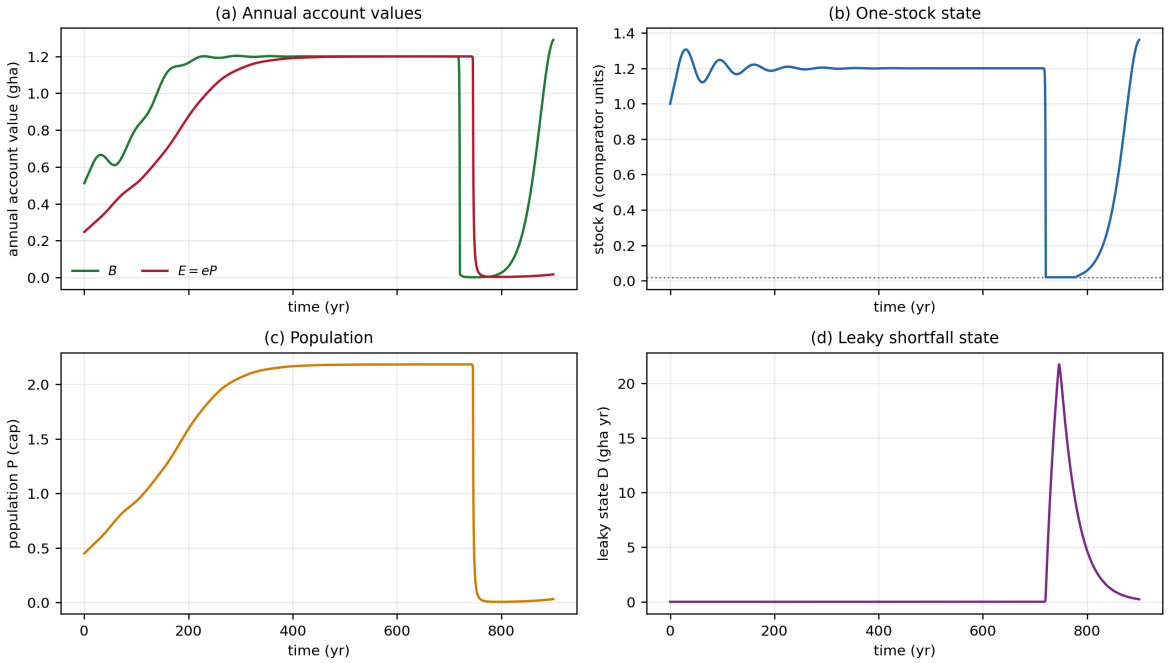


Figure S1: Full delayed one-stock comparator with a logistic technology increment. Parameters are $A_0 = 1.0$ stock unit, $P_0 = 0.45\,\mathrm{cap}$, $b_0 = 0.5$, $b_G = 0.6$, $\rho = 0.08\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\mathrm{gha\,cap}^{-1}$, $r = 0.02\,\mathrm{yr}^{-1}$, $\eta = 0.03\,\mathrm{yr}^{-1}$, $\alpha = 0.2\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\tau_g = 15\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-4}\,\mathrm{cap}$, $\Delta b = 0.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.04\,\mathrm{yr}^{-1}$, and $t_w = 120\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. The positive-part operator is exact. B and E are in gha, A is in comparator-specific stock units, P is in cap, and D is in gha yr. The trajectory reaches the imposed lower bound near 721 yr, then recovers as modeled demand falls; the bound is not absorbing. This is a conditional model illustration, not an observation or a two-land result.

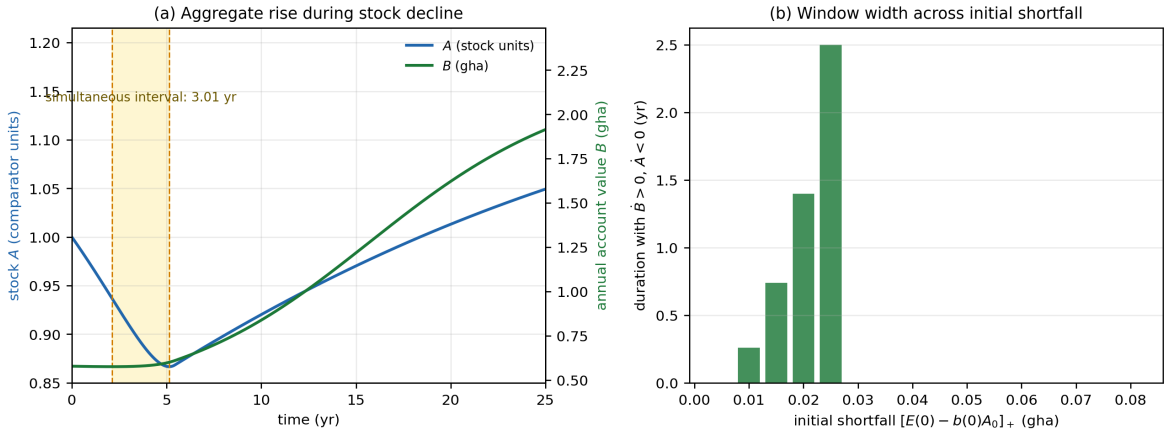


Figure S2: One-stock masking example with fixed demand and no delays. Parameters are $A_0 = 1.0$ stock unit, $E = 0.60\,\mathrm{gha}$, $b_0 = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $\eta = 0.05\,\mathrm{yr}^{-1}$, $\alpha = 0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\Delta b = 1.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.2\,\mathrm{yr}^{-1}$, and $t_w = 15\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. With the exact positive-part operator, the interval with $\dot{B} > 0$ and $\dot{A} < 0$ lasts about 3.0 yr. Over that interval, A falls from about 0.937 to 0.867 stock units while B rises from about 0.575 to 0.600 gha. Panel (b) shows the interval width across initial shortfalls $[E(0) - b(0)A_0]_+$ for this parameterization. These are conditional comparator outputs.

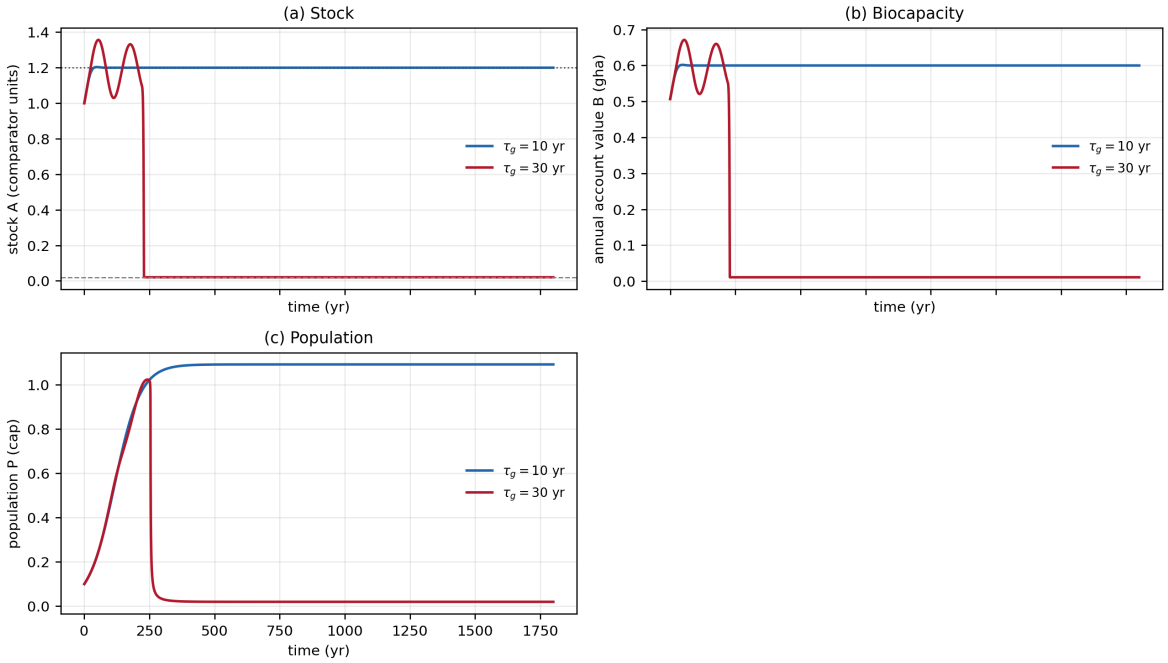


Figure S3: Controlled regeneration-delay examples in the debt-free one-stock subsystem. Both runs use $A_0 = 1.0$ stock unit, $P_0 = 0.1\,\mathrm{cap}$, $b = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\mathrm{gha\,cap}^{-1}$, $r = 0.02\,\mathrm{yr}^{-1}$, $\tau_p = 25\,\mathrm{yr}$, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-6}\,\mathrm{cap}$, and $\sigma = 1$; only τ_g differs (10 versus 30 yr). B is in gha, P in cap, and A in comparator-specific stock units. These selected trajectories do not establish a general critical delay.

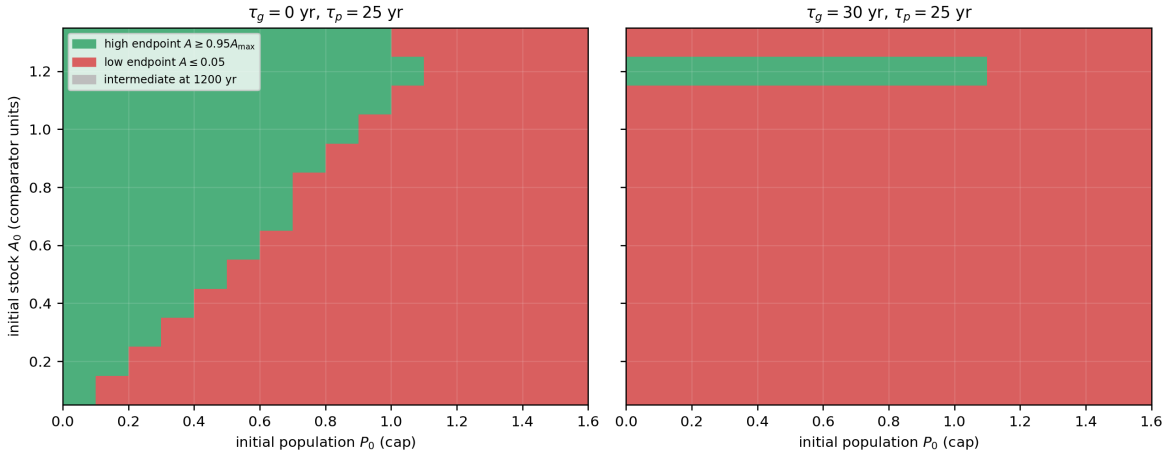


Figure S4: Finite-grid initial-condition classification in the debt-free one-stock subsystem. The runs use the parameter set in Fig. S3, with $K_{\min} = 10^{-6}\,\mathrm{cap}$ and $\tau_p = 25\,\mathrm{yr}$; $\tau_g = 0\,\mathrm{yr}$ in the left panel and 30 yr in the right. The imposed lower bound is $A_{\mathrm{ext}} = 0.02$ stock units. The grid uses $A_0 = 0.10, 0.20, \ldots, 1.30$ stock units and $P_0 = 0.05, 0.15, \ldots, 1.55\,\mathrm{cap}$; trajectories run to 1200 yr with $\Delta t = 0.5\,\mathrm{yr}$. Green marks a high endpoint $A(1200) \geq 0.95A_{\max}$, red a low endpoint $A(1200) \leq 0.05$ stock units, and gray an intermediate endpoint. These display-grid categories are not probabilities or continuous-basin measures.

S8 Controlled technology-wave experiment

The matched experiment compares endogenous-demand runs with the technology increment T_b switched on or off, at $P_0 = 1.0$ or $1.2\,\mathrm{cap}$. Within each pair, the runs share $A_{tot} = 2.5\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $A_{c0} = 0.9\,\mathrm{ha}$, $q_0 = 0.9$, and $D_0 = 0.0\,\mathrm{gha\,yr}$. Parameters and numerical methods are matched; only ω_{Tb} differs. Both runs also share constant pre-run histories at $q_0 = 0.9$ and the no-wave initial account value $B = 0.560184\,\mathrm{gha}$.

Shared parameter values are $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$, $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$, $b_{G,c} = 0.8\,\mathrm{gha\,yr\,ha^{-1}}$, $\rho_c = 0.08\,\mathrm{yr^{-1}}$, $\alpha = 0.03\,\mathrm{gha^{-1}\,yr^{-1}}$, $\eta = 0.05\,\mathrm{yr^{-1}}$, $\eta_f = 0.05\,\mathrm{yr^{-1}}$, $\mu = 0.06\,\mathrm{yr^{-1}}$, $e = 0.55\,\mathrm{gha\,cap^{-1}}$, $r = 0.02\,\mathrm{yr^{-1}}$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $\tau_{conv} = 1\,\mathrm{yr}$, and $\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$. The active wave uses $\Delta b = 1.0\,\mathrm{gha\,ha^{-1}}$, $\kappa_w = 0.1\,\mathrm{yr^{-1}}$, and $t_{wave} = 40\,\mathrm{yr}$; runs last $250\,\mathrm{yr}$. The logistic increment is already nonzero at $t = 0$ ($T_b(0) = 0.017986\,\mathrm{gha\,ha^{-1}}$), so initial B is $0.560184\,\mathrm{gha}$ without the wave and $0.570976\,\mathrm{gha}$ with it. This small difference comes from the logistic tail. Land, quality, population, and debt states otherwise match, and the pre-run B history remains common. The wave reaches half its increment at $t_{wave} = 40\,\mathrm{yr}$.

Figure S5 shows the $P_0 = 1.0$ pair. In both runs, peak debt occurs after the wave midpoint: $D_{max} = 0.008126\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ without the wave and $0.020205\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ with it. At $P_0 = 1.2$, the all-time maximum occurs early and is lower with the wave, but the maximum for $t \geq t_{wave}$ is higher. Table S13 reports both windows. Because t_{wave} marks the logistic midpoint, the post-midpoint window includes the wave's continued rise.

Table S13: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250\,\mathrm{yr}$; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

For $P_0 = 1.0$, the all-time wave/no-wave peak-debt ratio is 2.487 at $\Delta t = 0.05\,\mathrm{yr}$ and 2.492 at $0.025\,\mathrm{yr}$; under current-quality regeneration it is 2.422 at $0.05\,\mathrm{yr}$. For $P_0 = 1.2$, all-time ratios are 0.834 and 0.833, while post-midpoint ratios are 2.320 and 2.328. Halving the step changes debt maxima by less than 0.9%. In the $P_0 = 1.0$ pair, A_c stays at or above its initial $0.9\,\mathrm{ha}$, so this experiment illustrates a conditional delayed debt rebound, not the frozen-demand capital-land drawdown. At $P_0 = 1.2$, the all-time and post-midpoint maxima tell different stories. The debt variable D is a leaky state driven by aggregate deficit $E - B$, not an observed NFA or physical-stock quantity.

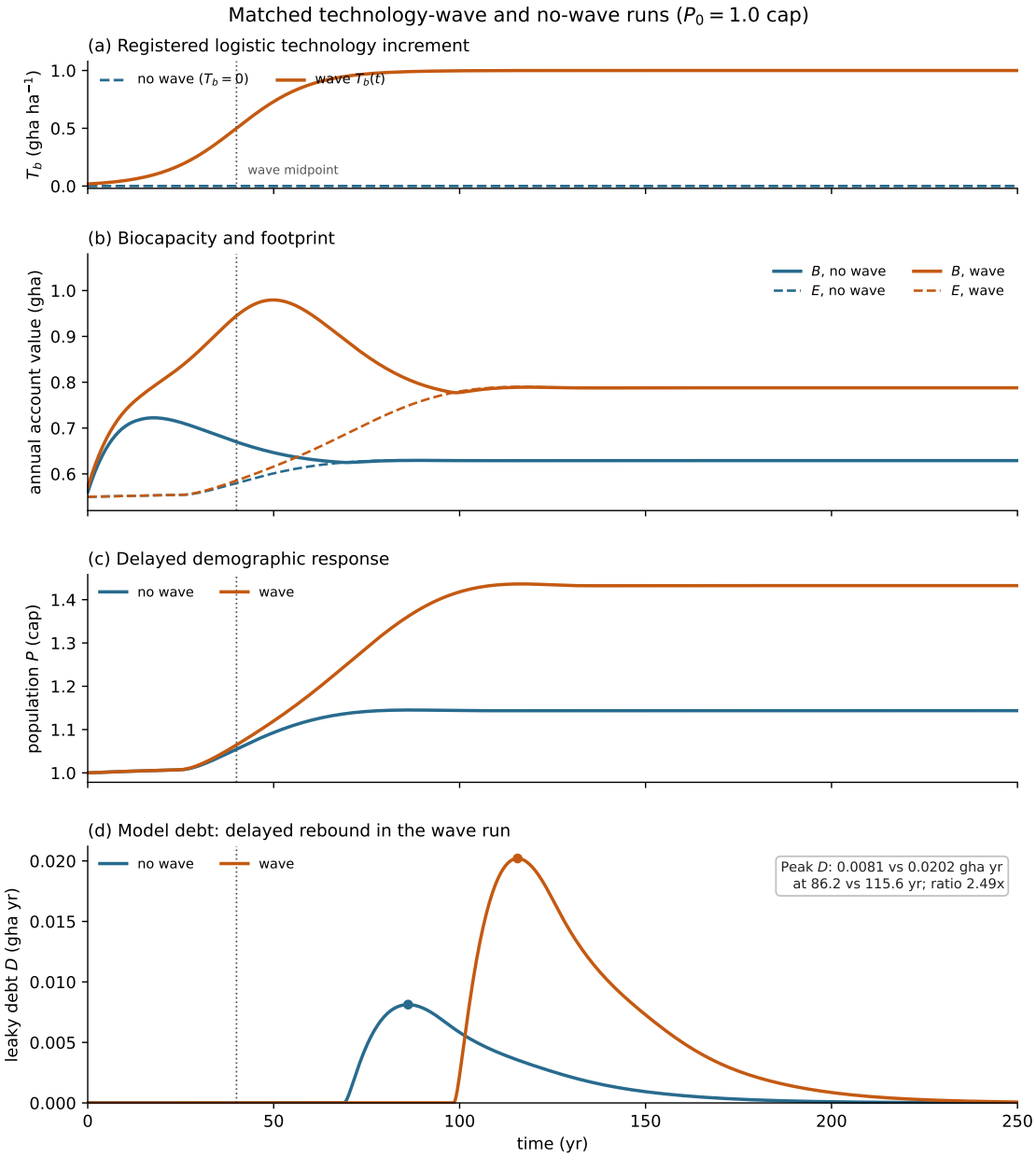


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); and (d) leaky model debt D (gha yr). Markers denote all-time maxima: 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}. The dotted line marks the midpoint, not an end date.

Supplementary references

References

Abaee, A. (2026a). *Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter*. Working paper (v41). https://github.com/MIKEAA2020/general-sustainability/blob/2efaa62083f2007dec3446c15c475af88f0925f3/arena%20agent%201/paper%20rewrites/latex/paper4_delay_dynamics_v41.pdf.

Abaee, A. (2026b). *Aggregation is a claim, not a presentation: the geometry of the acceptance gap between compensatory and noncompensatory transition assessment*. Working paper (v71). https://github.com/MIKEAA2020/general-sustainability/blob/2efaa62083f2007dec3446c15c475af88f0925f3/arena%20agent%201/paper%20rewrites/latex/paper06_assessment_separation_v71.pdf.

Global Footprint Network (2009). *Ecological Footprint Standards 2009*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/Ecological_Footprint_Standards_2009.pdf.

Global Footprint Network (2019). *Working Guidebook to the National Footprint Accounts 2019*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/National_Footprint_Accounts_Guidebook_2019.pdf.

Hutchings, J. A. & Reynolds, J. D. (2004). Marine fish population collapses: consequences for recovery and extinction risk. *BioScience*, 54(4), 297–309.

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M. & Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0*. Prepared for Footprint Data Foundation by researchers at York University and the University of Iceland. Workbook: https://footprint.info.yorku.ca/files/2026/04/NEFBA_Data_global_national_ecological_footprint_biocapacity.xlsx; release notes: https://footprint.info.yorku.ca/files/2026/04/NEFBA_ReleaseNotes.pdf; data page: https://footprint.info.yorku.ca/data/.

Neubauer, P., Jensen, O. P., Hutchings, J. A. & Baum, J. K. (2013). Resilience and recovery of overexploited marine populations. *Science*, 340(6130), 347–349.

Our World in Data (2026a). *Cereal yield*. FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cereal-yield.

Our World in Data (2026b). *Cropland area*. FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cropland-area.

Poeplau, C., et al. (2011). Temporal dynamics of soil organic carbon after land-use change in the temperate zone–carbon response functions as a model approach. *Global Change Biology*, 17(7), 2415–2427.

Poorter, L., et al. (2016). Biomass resilience of Neotropical secondary forests. *Nature*, 530(7590), 211–214.

United Nations et al. (2021). *System of Environmental-Economic Accounting—Ecosystem Accounting (SEEA EA)*. White cover publication, pre-edited text subject to official editing. New York: United Nations. https://seea.un.org/ecosystem-accounting.